

CONTINUOUS ASSESMENT TEST (CAT) – ACADEMIC YEAR 2024

STUDY GROUP: Computer Engineering & Computer Science

MODULE CODE: PHY1163, **MODULE NAME:** Physics for Engineers I

DATE: November 05, 2024, **DURATION:** 3 hours, **MAXIMUM MARKS** = 40

$$g = 9.81 \text{ m/s}^2$$

Question 1 / 15 marks

a) The air bubble formed by explosion inside water perform oscillations with time period T which depends on pressure (p), density (ρ) and on energy due to explosion (E). By using dimension analysis method, establish relation between T , p , E and ρ . 7 marks

b) A particle moves in a uniform circular motion with a radius of 3 cm. The net force radially acting on the object is 30 N. If the mass of the object is 6 kg,

i. What is centripetal acceleration? 2 marks

ii. What is the particle's linear speed? 2 marks

iii. What is the acceleration if the radius is doubled without changing the particle's speed? 2 marks

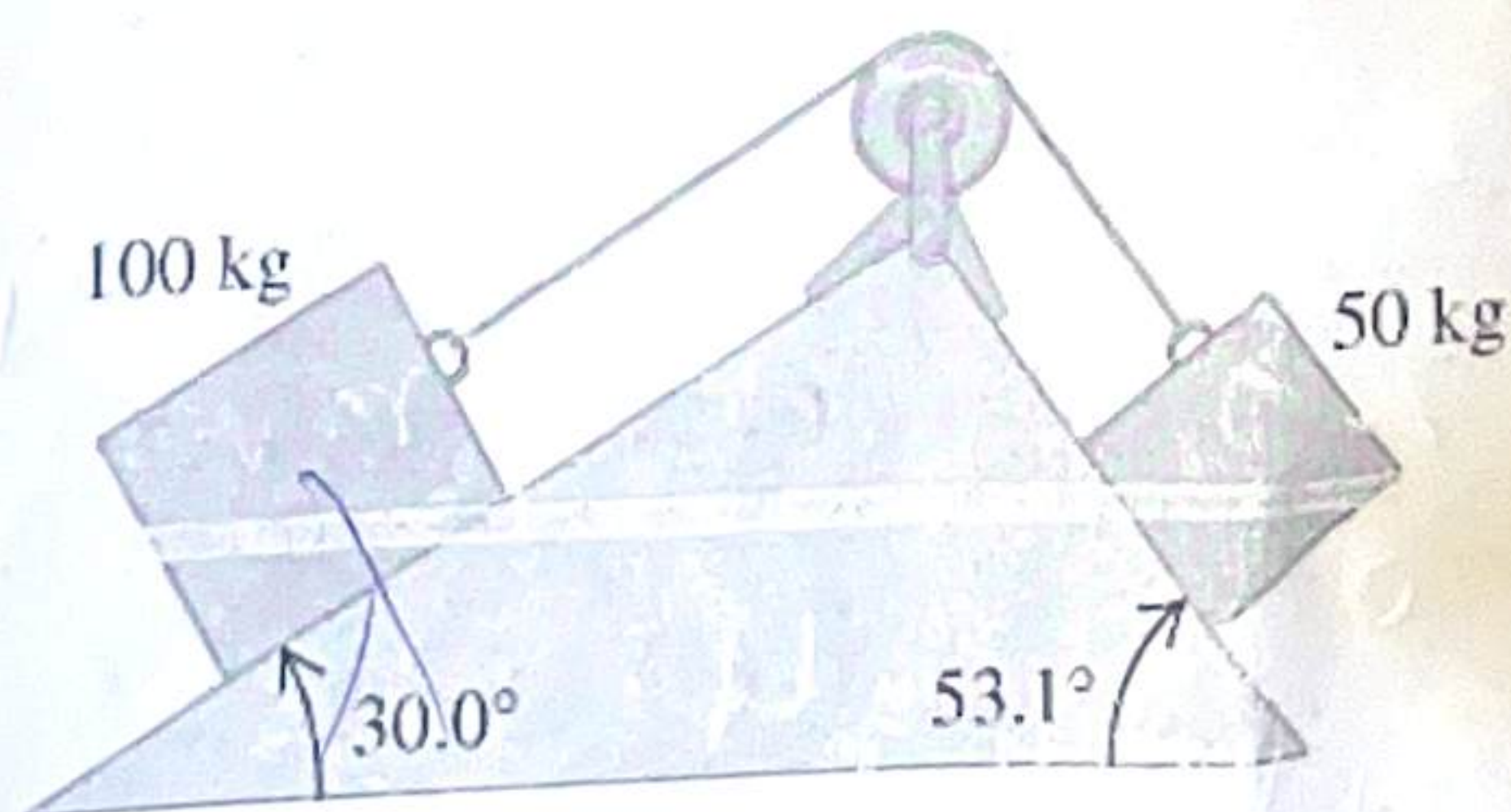
iv. What is the acceleration if the angular velocity is doubled without changing the circle's path? 2 marks

Question 2 / 15 marks

a) Two blocks connected by a cord passing over a small, frictionless pulley rest on frictionless planes

i. What is the acceleration of the blocks? 3 marks

ii. What is the tension in the cord? 3 marks



$$-100 \times 9.81 \times \sin 30 + 50 \times 9.81 \times \sin 53.1 = 150 a$$

$$-490.5 + 424.31 = 150 a$$

$$-66.19 = 150 a$$

$$a = -0.44 \text{ m/s}^2$$

Question 3 / 10 marks

I. A 5.00-kg ball, moving to the right at a velocity of +2.00 m/s on a frictionless table, collides head-on with a stationary 7.50-kg ball. Find the final velocities of the balls if the collision is

- Elastic and, 2 marks
- Perfectly Inelastic, 2 marks
- How much energy was lost in the inelastic collision? 2 marks

II. An object is launched at a velocity of 20 m/s in a direction making an angle of 25° upward with the horizontal.

- What is the maximum height reached by the object? 1 mark
- What is the total flight time of the object? 1 mark
- What is the horizontal range of the object? 1 mark
- What is the magnitude of the velocity of the object just before it hits the ground? 1 mark.

End of Questions